

CrossGaussian: Enhancing Remote Collaboration through 3D Gaussian Splatting and Real-time 360° Streaming

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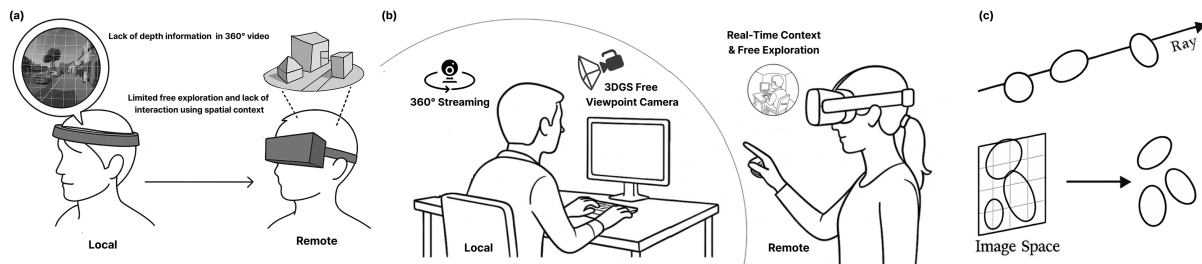


Figure 1: (a) 360° streaming limits spatial exploration and interaction due to missing depth. (b) CrossGaussian combines 360° streaming and 3DGS, exploring spatial visualization and interaction design to help remote users better understand context and interact effectively. (c) Gaussian Splatting enables fast, responsive 3D reconstruction of large environments.

Abstract

Remote users often face significant challenges in remote collaboration systems when joining virtual scenes reconstructed from a local user's environment. They are disadvantaged by the information asymmetry inherent in a shared virtual environment compared to local users. We present CrossGaussian, a VR collaboration system designed to address these limitations by providing remote users with a comprehensive 3D interactive view of the shared environment, created using 3DGS. It blends 360° video streaming and a large-scale 3DGS reconstruction via our automated pipeline. We then define the design space for visualization and interaction techniques that combine wide-coverage 3D and responsive 360° scenes.

CCS Concepts

• **Human-centered computing** → **Collaborative interaction.**

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Keywords

Remote Collaboration, 3D Gaussian Splatting, Virtual Reality

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1 Introduction

While co-located users can independently explore and interact with physical spaces, this autonomy is significantly constrained in remote collaboration [16]. Although 360° video streaming provides a wide field of view in remote environments, video-based scenes lack depth information and limit viewpoint control. While free engagement is critical for effective remote collaboration, expanding the coverage of shared environments remains challenging [13, 16]. Manually created 3D models enable free exploration [7, 12, 15] but are labor-intensive to create [6]. Recent advances in 3D reconstruction, such as photogrammetry-based progressive modeling using standard cameras, have enabled new approaches to remote

collaboration [4, 9]. However, its reliance on image-based Structure-from-Motion (SfM) produces surface-centric meshes with limited resolution, accuracy, and responsiveness. In contrast, Neural Radiance Fields (NeRF), a learning-based generative approach, provide high-fidelity reconstructions of complex structures [9, 16], but their high computational demands limit their practicality for large-scale environments. In this study, we focus on emerging methods in dynamic scene modeling enabled by 3D Gaussian Splatting (3DGS), which enables the fast, accurate, and highly responsive capture of large-scale physical environments [5, 10, 20, 22]. We present CrossGaussian, a system that blends real-time 360° video with 3D Gaussian Splatting (3DGS) to enable free-viewpoint exploration and novel viewing interactions in remote environments.

2 CrossGaussian

In 360° remote collaboration, remote users inevitably face several telepresence-related challenges: 1) a lack of spatial context in reconstructed environments, 2) limited access the local physical environment, and 3) spatiotemporal discrepancies between physical and virtual environments. Meanwhile, 3D Gaussian Splatting has enabled near real-time, GPU-accelerated rendering with editable 3D content [3, 11, 23]. We aim to address these challenges by introducing 3DGS into 360° video streaming and exploring the design space of visualization and interaction techniques.

2.1 System Architecture

To facilitate the efficient application of 3DGS in VR-based remote collaboration, we propose a fully automated pipeline that integrates preprocessing, remote model training, and rendering. By simply capturing a 360° video, users can initiate a process that runs entirely on remote GPU servers, enabling real-time 360° video streaming alongside the generation of high-fidelity 3DGS scenes.

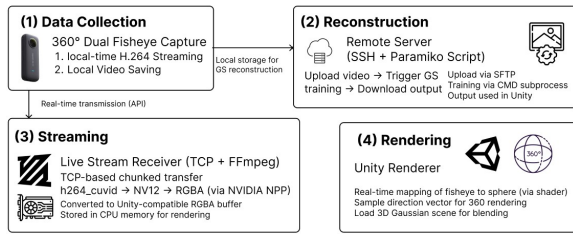


Figure 2: Automated Rendering Pipeline

The system has data collection, reconstruction, streaming, and rendering parts to construct remote environments [1, 8] (Figure 2).

2.2 Design & Implementation

Inspired by prior studies about scene blending in Cross-Reality [6, 16, 18, 19], we preemptively design the following core features for remote collaboration.

2.2.1 Blending of Overlapping Scenes. scene transitions can induce motion sickness and disrupt presence during scene changes [21]. To address this issue, our system overlaps a 3DGS scene with 360° video streaming in VR and applies visualization techniques, such

as transparency control [14, 17, 18] and color scaling [14]. In Figure 3-(a), this visual integration enables users to simultaneously perceive the real-time environmental context (360° stream) while freely exploring alternative viewpoints (3DGS), thereby reducing the cognitive load associated with context switching and maintaining presence. Leveraging this overlaid property, we apply color scaling to the 3DGS scene or adjust pixel values in the 360° video to perceptually separate non-salient regions (Figure 3-(b)). Gruenefeld et al. [6] enabled adjustable scene blending; similarly, our system allows users to control blending between 3DGS and 360° video for balanced contextual realism and interactive exploration.

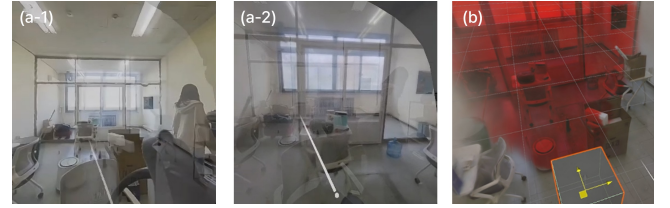


Figure 3: Layer blending of 360° streaming and 3DGS

2.2.2 Occlusion-Aware Exploration. 360° video lacks the ability to reveal areas occluded by structures due to the lack of depth information. We address this by leveraging view-dependent visibility data derived from a 3DGS model. The system automatically detect and highlight occluded regions that are not visible from the local user’s 360 camera perspective (Figure 4-(a)). Technically based on [2], We estimate per-pixel depth and pseudo-normals via neighboring pixel comparison, enabling precise shadow casting with Unity compute shaders and HLSL. Moreover, our system offers a see-through capability by using depth information available in a 3DGS model (Figure 4-(b)). Photogrammetry relies on a mesh-based representation with fixed surfaces, making it difficult to control transparency. In contrast, Gaussian Splatting uses 3D Gaussians with alpha values, allowing for natural semi-transparent rendering through alpha blending at the rendering stage. This enables more intuitive exploration and interaction without the need for complex viewpoint manipulation or information loss caused by occlusions. These two occlusion-aware visualization techniques unlock unique interaction capabilities by leveraging 3DGS in remote collaboration.



Figure 4: Occlusion-aware exploration via 3DGS depth

3 Perceptual Evaluation

We conduct a user study to examine how latency in reconstruction affects users’ perception of object existence and affordance. Even state-of-the-art 3D reconstruction often exceeds real-time limits (e.g., 33ms), yet prior work rarely examines how delays affect user

perception. To address this, we conducted a within-subject study using four randomized delay conditions (0.15 s, 1 s, 10 s, and 60 s). After observing an object under each condition, participants rated perceived manipulability trust in its existence on a 5-point Likert scale. They also reported subjective qualitative feedback.

We conducted the study with 18 participants. Due to violation in the normality assumption, we used the Friedman test with Wilcoxon signed-rank tests for post-hoc analysis. The results showed that delay in reconstruction significantly decreased perceived manipulability ($F = 27.22, p < .001$; 5.8 ± 1.6 (0.15 s), 5.7 ± 1.4 (1 s), 5.2 ± 1.4 (10 s), 4.3 ± 1.7 (60 s)). Particularly, we observed significant differences between 0.15 s-60 s, and 1 s-60 s conditions ($p < .05$). For perceived trust, the results was generally similar ($F = 29.87, p < .001$; 6.2 ± 1.2 (0.15 s), 5.8 ± 1.3 (1 s), 5.1 ± 1.5 (10 s), 4.3 ± 1.8 (60 s)). Along with 0.15 s-60 s, and 1 s-60 s conditions, 0.15 s-10 s was also significantly different ($p < .05$). Qualitative feedback revealed that most participants lost trust after 10 s, with 60 s delays making the object feel disconnected, reducing their willingness to interact.

4 Conclusion

CrossGaussian is the first to integrate a room-scale 3DGS reconstruction with 360° streaming, enabling view-dependent and post-processable scene representations for remote collaboration beyond object-scale approaches [9, 16].

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